

COMP3162 - Data Science Principles
Week 07 – Lab 06 (graded)

Data Cleaning and Preparation

Download and import the lab06 dataset posted on VLE.

Data Description:

The data records a snapshot of students' performance in few subjects. In addition, it records the reported income level of the household to which each student belong. The table below describes each feature in the dataset.

Feature Name	Description
ID	Unique identifier for each student
Eavg	Average of all scores earned for English Language over the period
Mavg	Average of all scores earned for Mathematics over the period
NumCoCurri	A count of the number of co-curricular activities in which the student participated
DaysAbsent	The number of sessions for which the student was absent for the school year. Each session is 0.5 days.
OverallAvg	The average of all subjects for the period, including Mathematics and English Language
H_inc	The income for the household in which the student resides

Clean, prepare and conduct the following exploratory analysis. Ensure that your dataset is reviewed after each modification to verify the expected outcomes.

NB. use notes in your rmarkdown file to state the results where outcomes are not visible.

1. Summary Statistics: find the structure, mean, min, max, draw histogram for all features in the dataset.
2. Determine the number of missing values in each feature. Based on your findings, decide on a strategy and apply this to the missing values (*ensure you write notes in your rmarkdown file to describe the decision taken*)
3. Remove the ID feature from the dataset.
4. Identify and fix outliers. Explain what was done.
5. Conduct Summary statistics on the modified dataset and discuss observations.
6. Descretize DaysAbsent to Low, Med, High

7. Construct a new field called Cparticipation which has True if the values in NumCoCurr is greater than 1, false otherwise
8. What percentage of students were absent more than 20% of the days in the school year?
9. How many students have an overall average below 50?
10. What percentage of students achieved each grade level for English Language, Mathematics and Overall Average, given the grade levels as follows:
A : 80 – 100
B : 70 – 79
C: 60 – 69
D: 50 – 59
F: below 50
11. Which subject did the students perform better at on aggregate?
12. What percentage of students who are in middle income households have an “A” average?